

Project Report

Asymmetric Encryption to Sign and Verify Data

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Cybersecurity Essentials Course
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September 11, 2026

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1 Introduction

This report documents the steps and learning outcomes from the hands-on lab titled *Asymmetric Encryption to Sign and Verify Data*, completed as part of the **Cybersecurity Essentials** course by the Office of the CISO Institute.

The objective of this lab was to provide practical experience in using asymmetric cryptography for creating and verifying digital signatures. Through this process, we ensured file integrity and authenticity, which are fundamental aspects of modern cybersecurity practices.

2 Objectives

The key learning objectives of this lab were:

- Generate asymmetric encryption keys using Google Cloud Key Management Service (KMS).
 - Create digital signatures for files using a private key stored securely in KMS.
 - Verify the integrity of files using a public key and OpenSSL.
 - Detect unauthorized modifications to files by validating digital signatures.
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3 Tools and Setup

The following tools and services were used in this lab:

- **Google Cloud Platform (GCP):** To securely create and manage cryptographic keys.
- **Google Cloud Shell:** A secure, browser-based Linux shell with pre-installed tools.
- **Google Cloud KMS (Key Management Service):** For managing private/public key pairs.
- **OpenSSL:** Used to verify signatures and perform cryptographic operations.

The environment was provided via a Qwiklabs temporary Google Cloud project with no cost incurred.

4 Step-by-Step Procedure

4.1 Step 1: Activate Google Cloud Shell

The lab started by activating Google Cloud Shell, which provided command-line access to the cloud project. Commands used:

```
gcloud auth list
gcloud config list project
```

This confirmed authentication and verified the correct project was selected.

4.2 Step 2: Create Asymmetric Key Pair

A KeyRing and an RSA key pair were generated to be used for signing files.

```
gcloud kms keyrings create lab-keyring --location us
```

```
gcloud kms keys create labkey \  
--location us \  
--keyring lab-keyring \  
--purpose "asymmetric-signing" \  
--default-algorithm "rsa-sign-pkcs1-4096-sha256"
```

This ensured that a secure private key was stored within the KMS while its corresponding public key could later be retrieved for verification.

4.3 Step 3: Prepare Files

Sample files were created and downloaded for signing.

```
echo "This is sample file 1" > file1.txt  
curl https://hadoop.apache.org/images/hadoop-logo.jpg > image1.jpg  
curl https://www.apache.org/images/asf_logo_wide.png > image2.jpg
```

4.4 Step 4: Generate Digital Signatures

Each file was signed using the private key stored in KMS. Example command for text file:

```
gcloud kms asymmetric-sign \  
--keyring lab-keyring \  
--key labkey \  
--version 1 \  
--location us \  
--input-file file1.txt \  
--signature-file file1.sig
```

Similarly, signatures were generated for image files.

4.5 Step 5: Retrieve Public Key

The public key was exported for use in signature verification.

```
gcloud kms keys versions get-public-key 1 \  
--key labkey \  
--keyring lab-keyring \  
--location us \  
--output-file key.pub
```

4.6 Step 6: Verify Signatures

OpenSSL was used to verify that the files had not been altered. Example verification command:

```
openssl dgst -verify key.pub -keyform PEM -sha256 \
-signature file1.sig -binary file1.txt
```

Output:

- **Verified OK** – file integrity confirmed.
- **Verification Failure** – file was altered or corrupted.

4.7 Step 7: Tampering Test

Files were intentionally altered to demonstrate detection:

1. The text file `file1.txt` was edited slightly, resulting in **Verification Failure**.
2. The image `image1.jpg` was replaced with another image, also resulting in **Verification Failure**.

Restoring the original files returned the verification result to **Verified OK**.

5 Lab Screenshot

Below is a placeholder for a screenshot showing the signing and verification process in the Google Cloud Shell:

```

100 14470 100 14470 0 0 58059 0 --i--- --i--- --i--- 57880
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ gcloud kms asymmetric-sign \
--keyring SKEYRING_NAME \
--key SKEY_NAME \
--version 1 \
--location SLOCATION \
--input-file file1.txt \
--signature-file file1.sig
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ gcloud kms asymmetric-sign \
--keyring SKEYRING_NAME \
--key SKEY_NAME \
--version 1 \
--location SLOCATION \
--input-file image1.jpg \
--signature-file image1.sig
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ gcloud kms keys versions get-public-key 1 \
--key SKEY_NAME \
--keyring SKEYRING_NAME \
--location SLOCATION \
--output-file key.pub
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ openssl dgst -verify key.pub -keyform PEM -sha256 -signature file1.sig -binary file1.txt
Verified OK
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ openssl dgst -verify key.pub -keyform PEM -sha256 -signature image1.sig -binary image1.jpg
Verified OK
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ openssl dgst -verify key.pub -keyform PEM -sha256 -signature file1.sig -binary file1.txt
Verification failure
40773841AC7D0000:error:02000068:rsa routines:ossl_rsa_verify:bad signature:../crypto/rsa/rsa_sign.c:430:
40773841AC7D0000:error:1C880004:Provider routines:rsa_verify:RSA lib:../providers/implementations/signature/rsa_sig.c:774:
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ openssl dgst -verify key.pub -keyform PEM -sha256 -signature file1.sig -binary file1.txt
Verified OK
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ mv image1.jpg image1.jpg-save
cp image2.jpg image1.jpg
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ openssl dgst -verify key.pub -keyform PEM -sha256 -signature image1.sig -binary image1.jpg
Verification failure
40E7B5A5F87B0000:error:02000068:rsa routines:ossl_rsa_verify:bad signature:../crypto/rsa/rsa_sign.c:430:
40E7B5A5F87B0000:error:1C880004:Provider routines:rsa_verify:RSA lib:../providers/implementations/signature/rsa_sig.c:774:
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $ cp image1.jpg-save image1.jpg
student_00_f70d3a3e4da3@cloudshell:~ (qwiklabs-gcp-04-d3d6f383b4fc) $

```

6 Skills and Experience Gained

Through this lab, the following skills and experience were gained:

Technical Skills

- Asymmetric cryptography using RSA and SHA-256.
- Digital signature creation and verification.
- Secure key management with Google Cloud KMS.
- Practical use of OpenSSL for security operations.
- Detecting file tampering and ensuring data integrity.

Experience

- Hands-on cloud-based cryptographic operations.
 - Real-world simulation of secure file distribution and verification.
 - Understanding PKI (Public Key Infrastructure) concepts and workflows.
 - Exposure to incident response by detecting unauthorized file modifications.
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7 Conclusion

This lab demonstrated the critical role of asymmetric encryption in ensuring data authenticity and integrity. By creating digital signatures with private keys and verifying them with public keys, it is possible to detect unauthorized modifications to files.

The skills developed in this exercise are directly applicable to real-world cybersecurity practices, including software integrity checks, secure communications, and incident response.

8 References

- Google Cloud Key Management Service Documentation: <https://cloud.google.com/kms/docs>
- OpenSSL Documentation: <https://www.openssl.org/docs/>
- Qwiklabs Cybersecurity Essentials Labs